

Baraq Lipshitz

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EDUCATION

ETH Zurich

Master of Science in Computer Science with Major in Machine Intelligence

Zurich, Switzerland

Sep 2024 – June 2026

New York University

Bachelor of Science in Computer Science

New York, NY

Aug 2021 – May 2024

Summa Cum Laude (highest honors) and graduated **1 year early**.

GPA: 3.99/4.0

SELECTED COURSEWORK

Algorithms (**Top 3 Students**), Large-Scale AI Engineering, Large Language Models, Advanced Systems Lab, Cloud Computing Architecture, Robot Vision, Probabilistic AI, Parallel Computing (CUDA, OpenMP, MPI)

PROJECTS

Mercedes-Benz Autonomous Driving & ETH Computer Vision Geometry group - Physically Consistent Video Generation Research

- Integrating V-JEPA 2 video representation learning with diffusion transformers (NVIDIA Cosmos) to improve temporal consistency and object dynamics for autonomous driving sim-to-real. Ongoing.

Research Project: Accelerating Sparse Ternary GEMM for Quantized LLM inference on Apple Silicon

- Optimized kernel for M-Series processors, achieving 5.98x speedup over TCSC baseline, sustaining 50.2% peak FLOPs/cycle throughput. Authored technical paper with methodology, ablations, and benchmarks.
- Developed novel data formats to boost memory locality, data compression, and reduce hot loop branching.

FSDP & Flash Attention for Llama 3 | PyTorch, SLURM

- Built a Llama 3 training pipeline for an NVIDIA GH200 cluster, enhanced with FSDP and FlashAttention.

Computer Vision 3D Maze Solver | vSLAM, VLAD, SIFT, OpenCV

- 1st place** in an NYU AI competition; led team to fastest virtual robot for 3D maze navigation [[video/code](#)].
- Designed a custom monocular depth estimation algorithm to reconstruct a 2D minimap, created a visual place recognition pipeline using SIFT, Bag of Visual Words with k-Means, and a Ball Tree data structure.

Spotify Song Genre Classification | XGBoost, PCA, Numpy, scikit-learn, Plotly

- End-to-end ML workflow on 50K-track dataset; statistically guided preprocessing, dimensionality reduction.
- Tuned XGBoost classifier with stratified k-fold CV; achieved 93.4% AUC across 10 genres [[write-up](#)].

WORK EXPERIENCE

Linvest21.ai - AI Engineer

May 2024 – Aug 2024 | New York, NY

- Led development of AlphaCopilot, an agentic AI research tool for wealth management offices.
- Designed a low-latency AWS database layer serving FRED series, real-time equity data, and news with vector similarity search. Built secure Python sandbox for LLM analysis and an SSE-connected React UI.
- Fast-tracked to leadership within weeks, 25% performance bonus. Reported to former JP Morgan CTO.

NYU High Speed Research Network - C++ Team Lead

Sep 2023 – May 2024 | New York, NY

- Led 3-person team to integrate Corelink into research projects (proprietary low-latency streaming platform).
- Planned development roadmap for a high-performance Unreal Engine 5 networking plugin, implementing scalable asynchronous data handling, and overcoming UE5.3+ restrictions on shared_ptrs.

Forbes Media - Technology Operations Intern

Jun 2023 – Aug 2023 | New York, NY

- Shipped a new forbes.com page estimated to drive **\$2.6M revenue** with React, Gatsby, Typescript [[link](#)].
- Supported CI/CD pipeline migration to custom Jenkins stack for APIs serving ~2 billion requests/month.

Payoneer - API Software Engineer Intern

Jun 2022 – Sep 2022 | New York, NY

- Worked on 7-person dev. team responsible for APIs handling **\$40B+** in annual international money transfers.
- Overhauled a high-throughput C#/NET webhook microservice and built SQL job queue, reducing latency 7x and improving dependent APIs used by Amazon, Google, and other enterprise clients.

Lightning Tech - Founder

Sep 2017 – Aug 2022 | Marin, CA

- Founded a consulting business in high school; \$30K+ profit across 200+ clients (businesses & consumers).